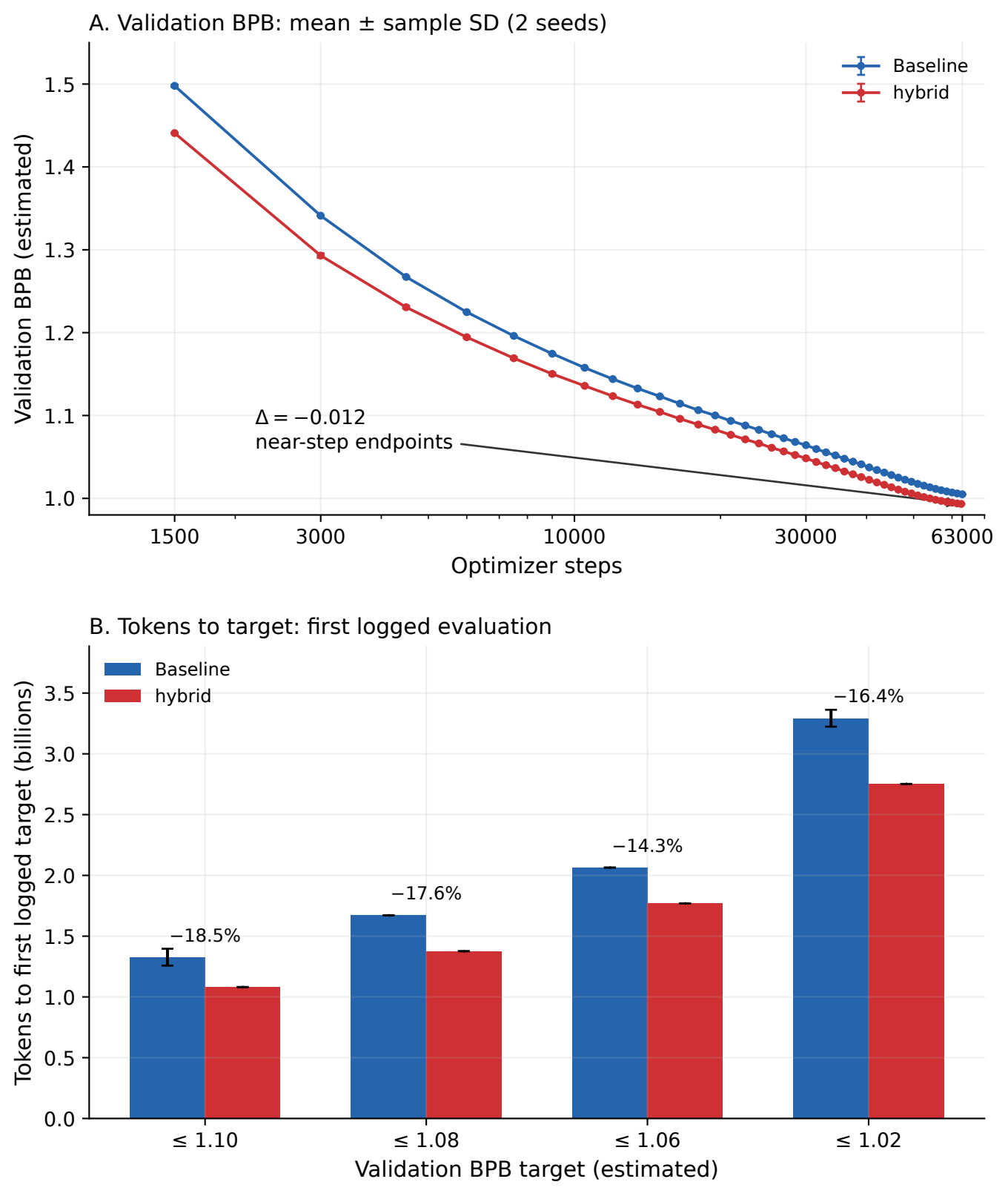
A. Validation BPB: mean ± sample SD (2 seeds)
Baseline
hybrid
Validation BPB (estimated)
1.5
1.4
1.3
1.2
1.1
1.0
Δ = −0.012
near-step endpoints
1500
3000
10000
30000
63000
Optimizer steps
B. Tokens to target: first logged evaluation
Baseline
hybrid
Tokens to first logged target (billions)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−18.5%
−17.6%
−14.3%
−16.4%
≤ 1.10
≤ 1.08
≤ 1.06
≤ 1.02
Validation BPB target (estimated)